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RESEARCH

Effectiveness of a novel low-level laser therapy protocol in managing breast cancer-related lymphedema

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Abstract

In this study, we investigate the effects of low-level laser therapy (LLLT) at different wavelengths on lymphedema to further determine the appropriate protocol and effectiveness of LLLT in patients with breast cancer-related lymphedema (BCRL). A randomized controlled trial was conducted involving 46 patients with BCRL. Participants were assigned to receive 904 nm LLLT, 650 nm LLLT, or sham treatment. ICG lymphography was used to assess dermal backflow as a marker of lymphatic dysfunction. Ultrasonography was used to assess subcutaneous thickness and tissue mechanical properties; shear wave velocity (SWV), with higher values indicating greater tissue stiffness, was measured using shear wave elastography. Extracellular fluid volume and local tissue water content were evaluated using multifrequency bioelectrical impedance analysis (MFBIA) and tissue dielectric constant (TDC), respectively. No significant between-group differences were observed in ICG lymphography, MFBIA, or TDC outcomes. For ultrasonographic outcomes, the overall between-group difference in the change in upper-lateral SWV ratio was significant ($p=0.029$), and Bonferroni-adjusted post-hoc analysis showed a greater reduction in Group A (904 nm) than in Group B (650 nm) (adjusted $p=0.026$). Within Group A, SWV ratios decreased significantly at the lower-lateral site and at the LLLT treatment site. These site-specific findings were not accompanied by measurable changes in lymphatic transport or fluid-related outcomes. Adding 904 nm–650 nm LLLT to conventional decongestive therapy did not produce significant additional effects on extracellular fluid-related or lymphatic functional outcomes. Site-specific reductions in SWV observed with 904 nm LLLT suggest a potential wavelength-dependent effect on local tissue mechanical properties; however, these findings are limited and should be considered exploratory. Larger adequately powered studies with prespecified primary outcomes and longer follow-up are required.

Keywords Lymphedema · Breast cancer · Low-level laser therapy · Photobiomodulation · Fibrosis · Randomized controlled trial

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Introduction

Low-level laser therapy (LLLT), also referred to as photobiomodulation (PBM), has been shown to reduce limb volume, relieve pain, and improve quality of life in patients with breast cancer related lymphedema (BCRL) after cancer related treatment [1, 2]. Experimental studies have also suggested several biological effects potentially relevant to lymphedema, including reduced fibrosis, enhanced lymphangiogenesis, improved lymphatic contractility, and decreased fluid accumulation [3–5]. These findings suggest that the therapeutic effects of LLLT may involve not only changes in lymphatic drainage and tissues.

The use of LLLT for treating lymphedema differs from earlier pain management laser treatments in terms of mechanism, wavelength, output power, and protocol, and thus requires distinct clinical application. LLLT induces biochemical changes at the cellular level by delivering low energy to the affected tissues [6, 7]. Currently, the only FDA-approved device for treating post-mastectomy lymphedema is the LTU-904 (RianCorp, Australia), which operates at a wavelength at 904 nm, with treated groups generally showing a reduction in limb volume [8]. Experimental evidence also supports antifibrotic effects at shorter near-infrared wavelength at 808 nm to reduced cololagen deposition and fibrosis in an animal model of muscle injury [9]. However, whether these wavelengths produce comparable effects in human lymphedematous tissue remains unclear, particularly because chronic lymphedema is characterized by progressive tissue remodeling and fibrosis. the effect of LLLT on skin fibrosis in lymphedema are lacking.

The uncertainty is clinically relevant because different therapeutic targets in BCRL may require different characteristics. Whereas PBM is known to be advantageous when targeting lymphatic structure under the skin layer, shorter wavelengths may produce a different distribution of energy within more superficial tissues. Thus, comparison of wavelengths with different optical characteristics may provide insight into whether the tissue response to LLLT is wavelength dependent. Nevertheless, direct clinical comparison of different wavelengths in patients with BCRL remain scarce, and the optimal wavelength and target tissue have not been established. The only clinical trial performed to date assessed the effect of LLLT on axillary lymph nodes and found its impact on distant skin fibrosis to be limited to transient effects [5]. Moreover, most clinical studies have evaluated a single wavelength and have focused on limb volume, circumference, or symptoms rather than directly assessing local tissue mechanical properties and lymphatic status.

Therefore, this randomized controlled trial aimed to compare 904 nm and 650 nm LLLT within sham treatment in patients with BCRL using multimodal objective assessments to further evaluate potential effects no tissue properties, fluid related measures, or lymphatic stagnancy.

Materials and methods

Patients

This prospective, single-blinded, parallel, randomized controlled trial was conducted at a tertiary hospital between December 2024 and January 2026. Eligible participants included patients diagnosed with secondary lymphedema following breast cancer surgery. Patients with primary lymphedema, a history of trauma, metastasis, or infection in both arms were excluded. Demographic and clinical characteristics, including age, body mass index, axillary lymph node dissection, chemotherapy, radiotherapy, time since lymphedema onset, and ISL stage, were recorded at baseline. Participants were randomly allocated into three groups: Group A (904 nm LLLT), Group B (650 nm LLLT), and a Group C (sham). Stratified randomization was used, accounting for prognostic factors such as lymphedema severity and participant age. Sample size was estimated based on the magnitude of the treatment effect reported in a previous systematic review and meta-analysis of LLLT for breast cancer-related lymphedema [10]. Smoot et al. reported a moderate-to-large between-group effect of LLLT on limb-volume reduction (standardized effect size, 0.62). Based on this evidence, a large overall treatment effect (Cohen's f = 0.50) was assumed for the three-group comparison. With a two-sided significance level of 0.05 and 80% power, the required sample size was estimated to be 42 participants (14 per group). Allowing for possible attrition, we aimed to recruit at least 15 participants per group. (Fig. 1) Block randomization with a fixed block size of four was performed using Microsoft Excel 2019. Random allocation sequences were generated with Excel's RAND function, organized into blocks to ensure equal distribution. Allocation concealment was maintained through sequentially numbered, opaque, sealed envelopes (SNOSE), which were opened only after baseline assessments. Participants and outcome assessors were blinded to group allocation. Outcome assessors were not involved in treatment sessions and had no access to treatment allocation. Blinding of the treating therapists was not feasible because the 904 nm and 650 nm interventions required different devices and device-specific operation. The sham intervention was administered under the same treatment conditions and for the same duration as active LLLT but without active

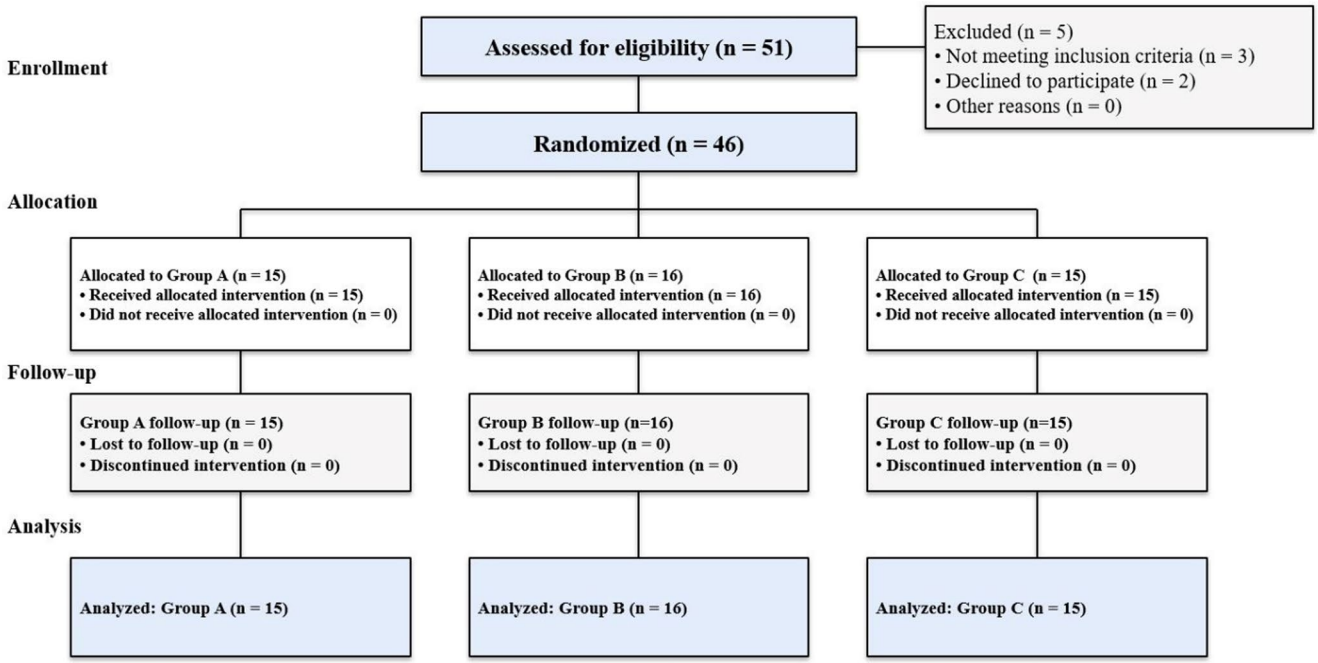


Fig. 1 CONSORT flow diagram of the randomized controlled trial

laser emission. All participants provided written informed consent, and the study was approved by the institutional review board (IRB No. D-2310-029-132. This study was prospectively registered at ClinicalTrials.gov (registration number: NCT07072780; https://register.clinicaltrials.gov/prs/beta/records, registration date: 23 November 2023.

Intervention

The overall treatment protocol is summarized in Fig. 2. All participants underwent a total of 10 treatment sessions, administered twice weekly over 5 weeks. Baseline assessments were performed immediately before the

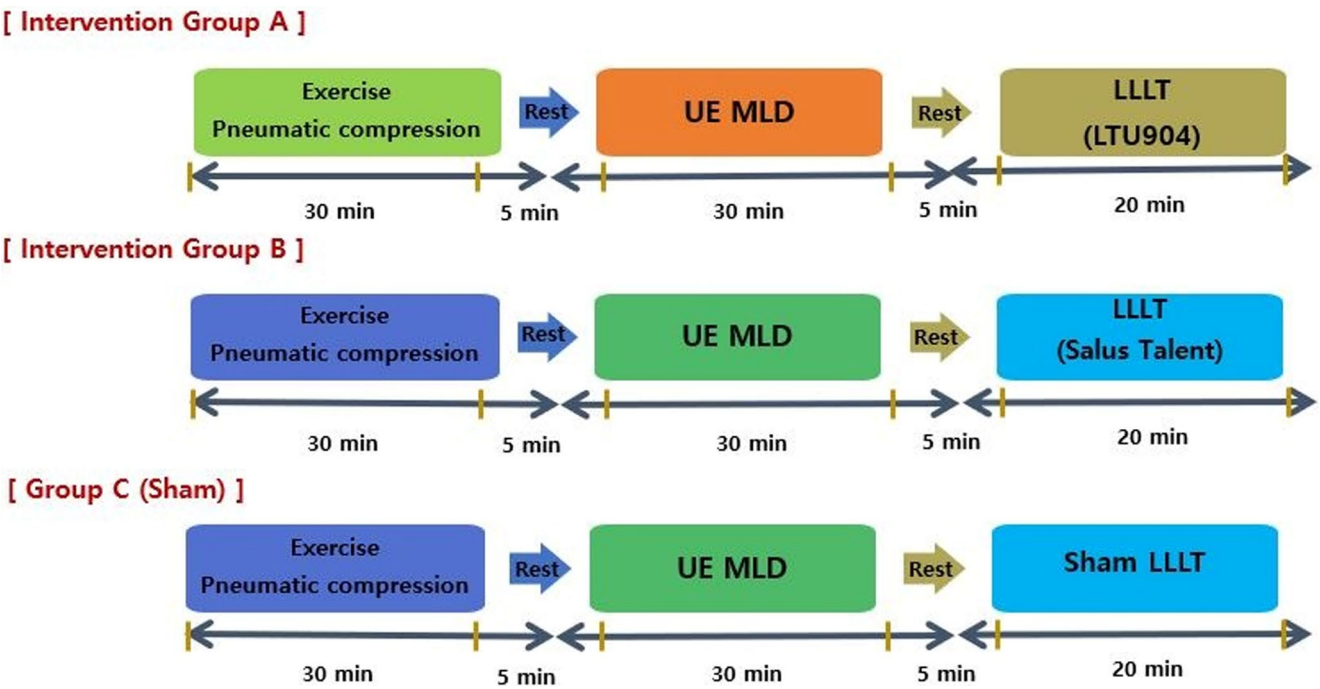


Fig. 2 A treatment protocol detailing the intervention and components of three groups

first treatment session, and post-intervention assessments were performed immediately after completion of the final treatment session. In every session, all three groups first received the same 60-minute conventional complex decongestive physiotherapy (CDPT), followed by 20 min of active or sham LLLT. CDPT consisted of manual lymphatic drainage (MLD), remedial exercises, and intermittent pneumatic compression and was administered by three licensed physical therapists trained in standard MLD techniques. The MLD component consisted of 30 min of fluoroscopy-guided MLD [11] targeting the affected upper limb. Thus, LLLT or sham treatment was evaluated as an adjunct to an otherwise identical CDPT program.

Following CDPT, participants were positioned supine with the dominant arm externally rotated at 90°. Group A received treatment using the LTU-904 (RianCorp, Australia) at a wavelength of 904 nm. Group B received the Salus Talent (Remed, Republic of Korea) device at a wavelength of 650 nm. The 904 nm wavelength was selected based on its established clinical use in BCRL, where LLLT has primarily been investigated for its effects on lymphatic drainage. The 650 nm wavelength was selected as a shorter-wavelength comparator with relatively shallower tissue penetration to explore whether wavelength-dependent responses differed when targeting chronic skin and subcutaneous tissue changes associated with BCRL. Both devices were commercially available PBM systems used with their manufacturer defined clinical treatment settings. Because the two devices differed in wavelength and emission mode, their output parameters were not independently modified to create physically identical irradiation conditions. Device-specific parameters, including emission mode, peak and average output power, irradiance, pulse frequency, duty cycle, beam or spot area, and exposure duration, are summarized in Supplementary Table 1 [12]. Group C received placebo laser therapy with no active laser emission. LLLT was applied to a circular area (7 cm diameter) selected individually based on ICG lymphography findings rather than predefined anatomical regions. Specifically, the target area corresponded to the region demonstrating the most prominent dermal backflow (DB) pattern on ICG lymphography, which represented the area with local fibrosis and the greatest local edema due to lymphatic dysfunction. For the LTU-904, the treatment area was adjusted at 1 cm intervals, with each point irradiated for one minute (Fig. 3, Supplementary Video 1). Five-minute rest intervals were included between each intervention. Pre- and post-intervention assessments are described in detail in the following sections. Participants and outcome assessors were blinded to group allocation. All outcome assessments were performed by assessors who were independent of the treating therapists and were not involved in or present during the treatment sessions.



Fig. 3 The application of LLLT (LTU-904) at a targeted dermal backflow site

Outcome measures

ICG lymphography

ICG lymphography was conducted before and after the intervention using a near-infrared camera system (Lymphoscope, Busan, Korea) (Fig. 3) [13, 14]. Indocyanine green (Diagnogreen 2.5 mg/mL) was injected intradermally into the first and third web spaces of the affected hand (0.1 mL each). To minimize injection discomfort, lidocaine HCl 2% with epinephrine (1:100,000) was administered beforehand. The same injection and imaging protocol was applied to all treatment groups at both assessments, thereby minimizing differential effects on between-group comparisons. Images were classified from Type I to V based on the dermal backflow (DB) stage, reflecting the spread and pattern of lymphatic stasis [15, 16]. Segmental DB staging was also performed to assess localized lymphatic status in five regions: upper lateral and medial arm, lower lateral and medial upper arm, and dorsum of the hand [17].

Ultrasonography

Subcutaneous tissue thickness was measured at five predefined sites of the affected upper limb using B-mode ultrasonography (7.5 MHz linear-array transducer, Siemens Medical Solutions). Landmarks included the midpoint of the wrist crease, medial and lateral epicondyles, and biceps tendon. From the elbow midpoint, measurements were taken 10 cm proximally and distally along predefined lines on the medial and lateral sides [18]. Thickness between the first and second fingers was also assessed. Adequate gel was applied to avoid compression artifacts. Resistance to compression (RC) was calculated by applying maximal compression and expressing the

change in thickness as a fraction of the initial value [19]. Shear wave elastography was performed using a curved transducer, with shear wave velocity (SWV, m/s) reflecting tissue stiffness at the region of interest (ROI). SWV and RC values were normalized to the contralateral side to produce ratios [20]. All measurements were performed in triplicate by a single physiatrist, and mean values were used for analysis (Fig. 4, A-C).

Multiple Frequency Bioelectrical Impedance Analysis (MFBIA)

Extracellular fluid volume was measured using the S10 device (InBody, Seoul, Korea). The limb index ratio (LIR) was calculated by comparing impedance between affected and unaffected arms at 1 and 5 kHz. Zero-frequency impedance (R0), which reflects extracellular water (ECW), was also measured. The R0 ratio (R0 L/U) was derived by comparing upper and lower extremity impedance on the affected side [21, 22].

Tissue Dielectric Constant (TDC)

Local tissue water content was assessed using the Lymphscanner (Delfin Technologies, Kuopio, Finland). The device measures localized percent water content (PWC) via electromagnetic signals. A handheld probe was applied for approximately 10 s at six standardized sites on both arms. Measurement sites included: TDC1 (ventral forearm, 15 cm below elbow crease), TDC2 (medial upper arm, 3 cm above the medial epicondyle), TDC3 (ventral upper arm, 7 cm above elbow crease), TDC4 (hand dorsum between thumb

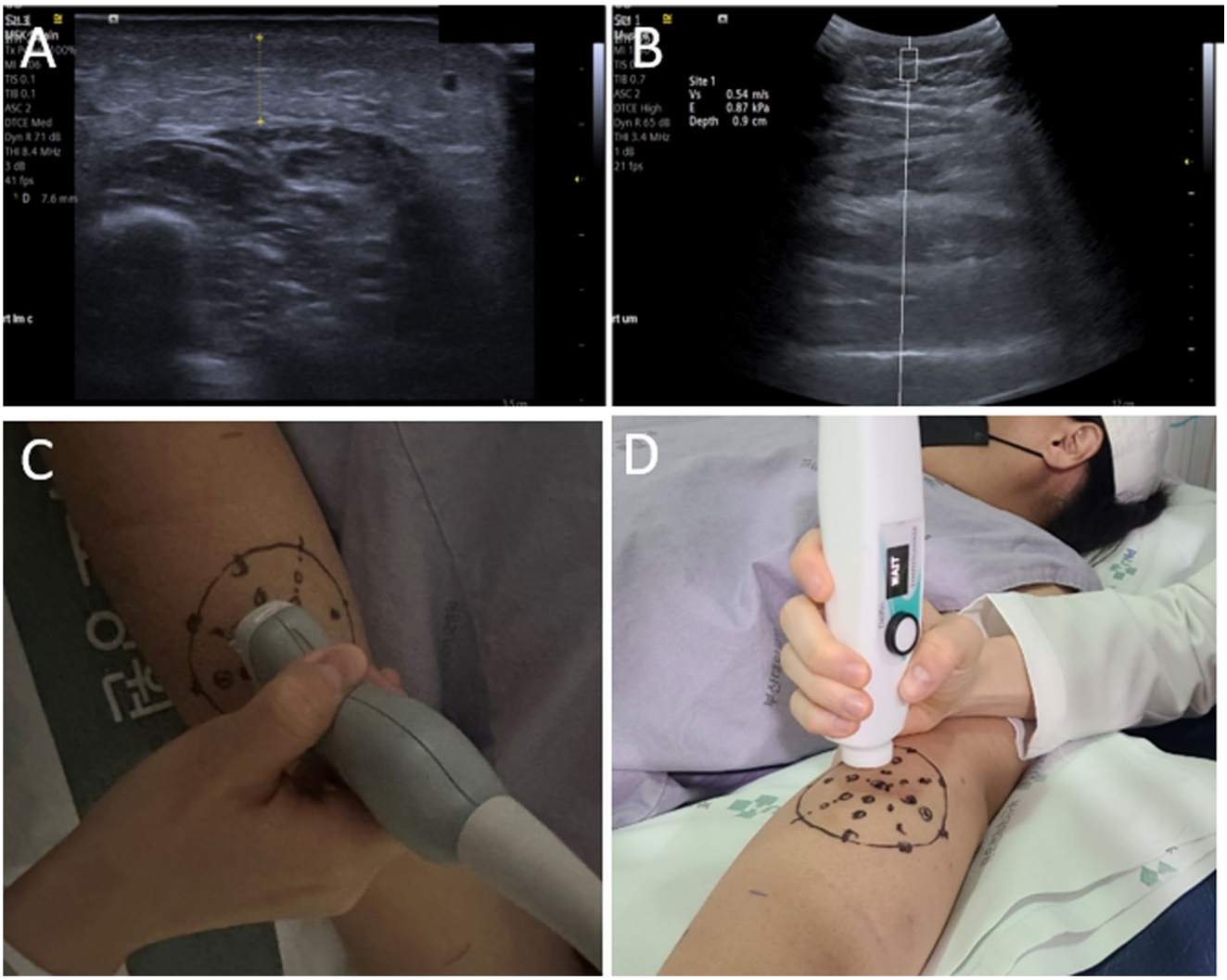


Fig. 4 Measurement methods: (**A**) ultrasonographic subcutaneous thickness, (**B**) shear wave velocity mapping, (**C**) measuring elastography at the treatment site, and (**D**) measuring tissue dielectric constant (TDC) at the treatment site

and index), TDC5 (dorsal forearm, 10 cm below olecranon), and TDC6 (dorsal upper arm, 7 cm above olecranon). Three repeated measurements were averaged for each site (Fig. 4, D) [5, 23].

Statistical analysis

Continuous variables are presented as mean ± standard deviation (SD) or median with interquartile range (IQR), depending on the normality of distribution. For comparisons among the three groups, one-way ANOVA or the Kruskal-Wallis test was applied, depending on the results of the Shapiro–Wilk test for normality. Categorical variables were compared using the chi-squared test or Fisher's exact test, as appropriate. Post-hoc analysis for inter-group comparisons was conducted using the Bonferroni correction. For within-group comparisons before and after intervention, the paired t-test or Wilcoxon signed-rank test was employed, depending on data normality. All statistical analyses were conducted using R version 4.4.2 (R Core Team, 2024; http://cran.r-project.org). A p-value < 0.05 was considered statis tically significant.

Result

Baseline characteristics did not differ significantly among the three groups (Table 1). ICG lymphography showed no significant between-group differences in changes in segmental DB stage after the intervention. Similarly, neither MFBIA nor TDC demonstrated significant between-group changes in extracellular fluid volume or local tissue water content. Among ultrasonographic outcomes, overall group differences were observed for the upper-medial subcutaneous thickness ratio and RC ratio, but post-hoc comparisons were not significant after Bonferroni correction. The change in upper-lateral SWV ratio differed significantly among groups (overall $p = 0.029$); Bonferroni-adjusted post-hoc analysis showed a greater reduction in Group A than in Group B (adjusted $p = 0.026$) (Fig. 5). Within-group analyses showed significant reductions in SWV ratio in Group A at the lower-lateral site and at the LLLT treatment site (Fig. 6). No significant within-group changes were observed in MFBIA, TDC, or ICG lymphography outcomes (Tables 2 and 3). These SWV findings were site-specific and were not accompanied by significant changes in fluid-related or lymphatic functional measures.

Discussion

The principal finding of this study was that neither 904 nm–650 nm LLLT produced significant additional changes in fluid-related or lymphatic imaging outcomes compared with sham treatment. In contrast, a site-specific change in tissue mechanical properties was observed especially to individually selected treatment sites in the 904 nm group. These findings suggest a potential tissue-specific

Table 1 Baseline characteristics of the participants

Characteristic	Group A ($n = 15$)	Group B ($n = 16$)	Group C ($n = 15$)	P value
Age (yr), mean (SD)	56.47 (11.83)	58.25 (10.82)	55.00 (9.33)	0.701
BMI, mean (SD)	22.48 (2.29)	23.68 (3.37)	23.08 (2.90)	0.235
ALD (%)	7 (46.7)	7 (43.8)	10 (66.7)	0.387
Chemotherapy (%)	13 (83.7)	12 (75.0)	13 (86.7)	0.610
Radiotherapy (%)	14 (93.3)	9 (56.2)	10 (66.7)	0.063
Onset (yr), mean (SD)	4.53 (2.70)	5.25 (3.38)	4.07 (2.94)	0.551
ISL stage (%)				
Stage 1	0 (0.0)	1 (6.2)	2 (13.3)	0.308
Stage 2	11 (73.3)	8 (50.0)	9 (60.0)	
Stage 3	2 (13.3)	3 (18.8)	4 (26.7)	
Stage 4	2 (13.3)	4 (25.0)	0 (0)	
Baseline MFBIA values, mean (SD)				
R0 Lo/Up	0.85 (2.70)	0.79 (0.25)	0.71 (0.11)	0.342
1 Hz LIR	1.24 (0.19)	1.20 (0.27)	1.20 (0.23)	0.880
5 Hz LIR	1.22 (0.23)	1.20 (0.28)	1.19 (0.23)	0.703
Baseline PWC values LLLT site, mean (SD)	1.08 (0.15)	1.07 (0.16)	1.08 (0.20)	0.944

Data are presented as mean ± standard deviation (SD) or number (%). Continuous variables were compared using one-way analysis of variance (ANOVA) or the Kruskal–Wallis test according to normality assessed via the Shapiro–Wilk test. Categorical variables were compared using the chi-squared test or Fisher's exact test, as appropriate. Statistical significance was defined as $p < 0.05$

BMI body mass index, *MFBIA* multifrequency bioelectrical impedance analysis, *Lo/Up* Lower/Upper extremity, *LIR* limb index ratio, *PWC* percentage of water content, *LLLT* low-level laser therapy

*$p < 0.05$

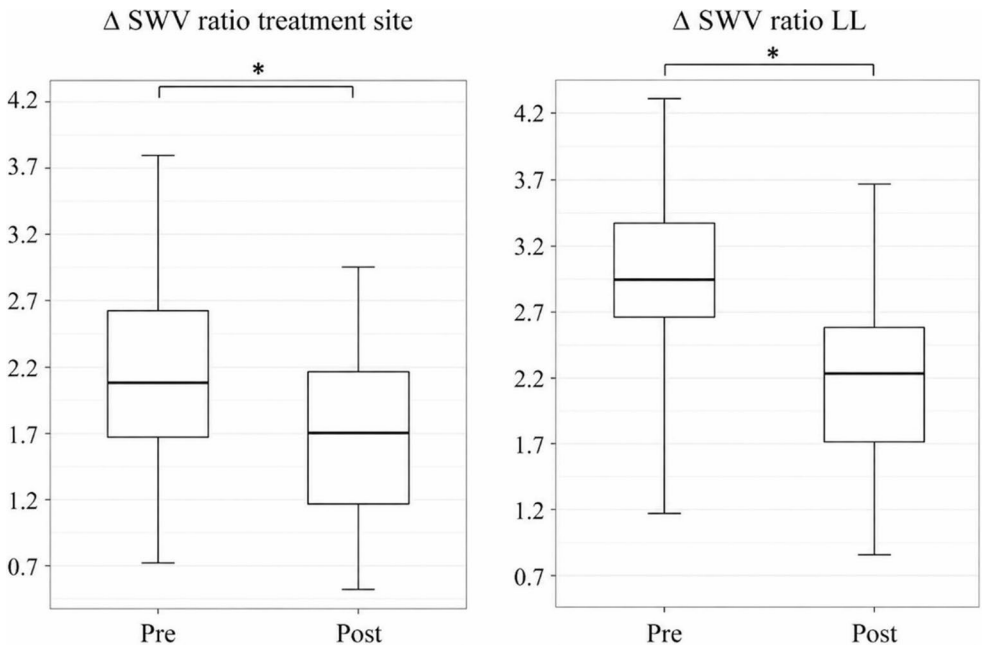


Fig. 5 An intra-group analysis showing the improvement of pre- and post-treatment changes in SWV in Group A

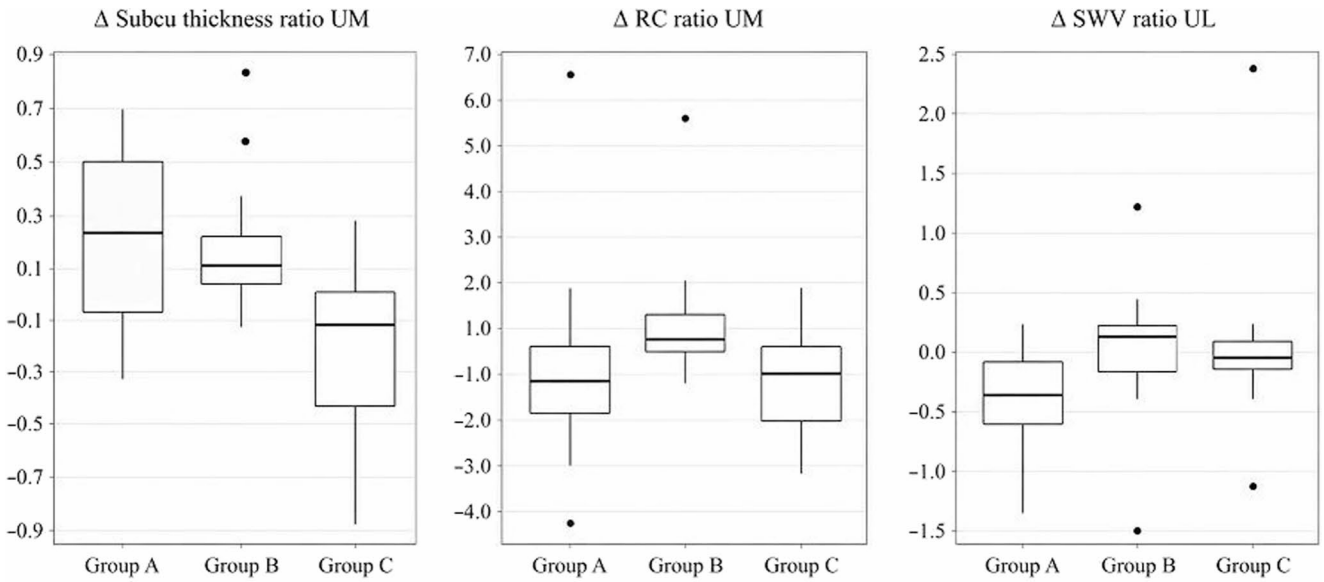


Fig. 6 An inter-group analysis of ultrasonographic parameters: subcutaneous thickness and SWV

response that warrants further investigation. Importantly, these findings should be interpreted in the context of the treatment target used in the present study. Previous clinical LLLT protocols for BCRL have primarily aimed to facilitate lymphatic drainage, with irradiation commonly directed toward proximal lymphatic structures or the axillary region and treatment effects evaluated mainly by reductions in limb volume or circumference. In contrast, the present study deliberately modified this conventional treatment strategy by directing LLLT toward individualized peripheral regions with prominent dermal backflow and chronic tissue changes identified by ICG lymphography. Therefore, the absence of significant changes in fluid-related or lymphatic outcomes in the present study does not necessarily contradict previous reports of edema reduction, because the anatomical target and intended tissue response differed substantially.

All participants in the present study received the same standardized CDPT regimen because LLLT was evaluated as an adjunct to conventional lymphedema treatment rather than as a stand-alone therapy. This design was intended to reflect clinical practice and to determine whether adding LLLT at different wavelengths provides additional benefits beyond those achieved with CDPT alone. Accordingly, improvements observed similarly across all groups cannot

Table 2 Inter-group analysis of differences in clinic-functional values after treatment

Characteristic	Group A ($n=15$)	Group B ($n=16$)	Group C ($n=15$)	P value
MFBIA values, median (IQR)				
Δ R0 Lo/Up	-0.01[-0.05, 0.03]	0.00 [-0.03, 0.05]	0.02 [0.00, 0.04]	0.477
Δ 1 Hz LIR	0.02 [-0.01, 0.08]	-0.02 [-0.04, 0.02]	0.00 [-0.05, 0.03]	0.189
Δ 5 Hz LIR	0.03 [-0.03, 0.07]	0.00 [-0.04, 0.02]	0.00 [-0.05, 0.03]	0.437
Ultrasonography values, median (IQR)				
Δ subcu thickness ratio UL	0.07 [-0.21, 0.13]	-0.03 [-0.19, 0.24]	-0.02 [-0.27, 0.17]	0.769
Δ subcu thickness ratio UM	0.25 [-0.09, 0.57]	0.11 [0.03, 0.25]	-0.07 [-0.46, 0.05]	0.030*
Δ subcu thickness ratio LL	0.17 [0.11, 0.33]	0.10 [-0.06, 0.32]	0.07 [-0.12, 0.34]	0.849
Δ subcu thickness ratio LM	-0.42 [-0.63, 0.30]	-0.08 [-0.41, 0.30]	0.05 [-0.24, 0.22]	0.642
Δ subcu thickness ratio hand	-0.05 [-0.23, 0.33]	0.01 [-0.38, 0.05]	0.17 [-0.17, 0.28]	0.430
Δ RC ratio UL	-0.14 [-0.65, 0.03]	-0.26 [-0.46, 0.81]	-0.18 [-0.63, 0.15]	0.796
Δ RC ratio UM	-0.66 [-1.17, 0.29]	-0.35 [-1.49, 0.26]	0.39 [0.15, 0.94]	0.033*
Δ RC ratio LL	-0.05 [-0.28, 0.44]	0.27 [-0.12, 0.67]	-0.22 [-0.40, 0.57]	0.798
Δ RC ratio LM	0.08 [-0.51, 0.77]	0.14 [-0.26, 0.52]	0.04 [-0.53, 0.40]	0.776
ΔRC ratio hand	-0.11 [-0.40, 0.16]	0.10 [-0.68, 0.61]	0.45 [-0.76, 1.74]	0.409
Δ SWV ratio UL	-0.43 [-0.59, -0.06]	0.14 [-0.20, 0.27]	-0.01 [-0.17, 0.16]	0.029*
Δ SWV ratio UM	0.05 [-0.23, 0.33]	-0.02 [-0.37, 0.16]	0.08 [-0.17, 0.26]	0.517
Δ SWV ratio LL	-0.33 [-0.94, 0.04]	-0.09 [-0.36, 0.20]	-0.22 [-0.47, 0.35]	0.506
Δ SWV ratio LM	-0.37 [-0.43, 0.13]	0.00 [-0.23, 0.21]	-0.02 [-0.24, 0.28]	0.065
Δ SWV ratio LLLT site	-0.25 [-1.07, 0.12]	-0.13 [-0.44, 0.07]	-0.26 [-0.94, 0.13]	0.646
ICG lymphography DB stage, median (IQR)				
ΔDB stage UL	0.00 [0.00, 0.75]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.209
Δ DB stage UM	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.466
Δ DB stage LL	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.212
Δ DB stage LM	0.00 [0.00, 0.75]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.150
Δ DB stage Hand	0.00 [0.00, 0.75]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.119

Data are presented as median [interquartile range (IQR)] unless otherwise indicated. Changes (Δ) were calculated as post-intervention values minus pre-intervention values. Inter-group comparisons were performed using one-way analysis of variance (ANOVA) or the Kruskal–Wallis test according to normality assessed via the Shapiro–Wilk test. Post hoc pairwise comparisons were performed using Bonferroni correction when overall group differences were statistically significant. Post-hoc pairwise comparisons were performed using Bonferroni correction when overall group differences were statistically significant. For the upper-lateral SWV ratio, the overall group comparison was $p=0.029$ and the Bonferroni-adjusted Group A versus Group B comparison was $p=0.026$. Statistical significance was defined as $p<0.05$

MFBIA multifrequency bioelectrical impedance analysis, *Lo/Up* lower/upper extremity, *LIR* limb index ratio, *UL* upper lateral, *UM* upper medial, *LL* lower lateral, *LM* lower medial, *RC* resistance to compression, *SWV* shear wave velocity, *LLLT* low-level laser therapy, *ICG* indocyanine green, *DB* dermal back flow

be attributed specifically to LLLT and may reflect the effects of CDPT. The absence of significant between-group differences in MFBIA, TDC, and ICG lymphography therefore suggests that, under the present 5-week protocol, neither 904 nm–650 nm LLLT produced a measurable additional effect on tissue fluid or lymphatic imaging outcomes beyond standard CDPT.

There are various factors that determine the extent to which the energy emitted from LLLT is absorbed by human tissue, among which the wavelength of the laser determines the extent of tissue penetration and the energy absorbed and scattered into the tissue. In general, the longer the wavelength, the deeper it penetrates the tissue. The range for reaching the target tissue and producing a therapeutic effect is the infrared and near-infrared wavelength of 650–905 nm, which is generally known to be the most effective range for the skin, subcutaneous tissue, and other areas targeted by LLLT. Otherwise, the results of the previous study on biological tissue demonstrated that LLLT according to wavelength has different attenuation and transmission power depending on the anatomical structure [24]. Since each wavelength has different tissue penetration depth and biological action mechanism, it is necessary to establish a basis for selecting an appropriate wavelength depending on the patient's symptoms and status of tissue after lymphedema. Considering the result of our study, there was a significant improvement in the subcutaneous thickness of the medial upper side of the limb in the group A compared to the other two groups after treatment. Otherwise, in post hoc analysis, this did not show a significance with the group B and group C. In addition, PWC through TDC measurement did not show a significant difference between the groups after treatment therefore, it is inconclusive to confirm that the current variable delivers sufficient wavelength and energy to

Table 3 Intra-group analysis of differences in clinic-functional values after treatment

	Group A ($N=15$)				Group B ($N=16$)				Group C ($N=15$)			
Characteristic	Pre	Post	Differences (95% CI)	*P* value	Pre	Post	Differences (95% CI)	*P* value	Pre	Post	Differences (95% CI)	*P* value
MFBIA values, median (IQR)												
Δ R0 Lo/Up	0.85	0.86	-0.01 (-0.054, 0.029)	0.599	0.84	0.81	0.03 (-0.059, 0.129)	0.952	0.75	0.73	0.02 (-0.028, 0.065)	0.327
Δ 1 Hz LIR	1.24	1.21	0.03 (-0.011, 0.069)	0.107	1.18	1.22	-0.05 (-0.157, 0.062)	0.426	1.15	1.13	0.02 (-0.041, 0.085)	0.410
Δ 5 Hz LIR	1.22	1.21	0.01 (-0.057, 0.071)	0.330	1.17	1.22	-0.05 (-0.168, 0.069)	0.715	1.14	1.10	0.05 (-0.048, 0.141)	0.327
Ultrasonography values, median (IQR)												
Δ subcu ratio UL	1.13	1.14	-0.01 (-0.203, 0.188)	0.946	1.30	1.24	0.06 (-0.171, 0.301)	0.635	1.25	1.32	-0.07 (-0.249, 0.111)	0.263
Δ subcu ratio UM	1.50	1.27	0.23 (-0.020, 0.480)	0.094	1.51	1.32	0.18 (0.007, 0.361)	0.040*	1.24	1.42	-0.18 (-0.398, 0.033)	0.104
Δ subcu ratio LL	1.23	1.09	0.14 (-0.053, 0.331)	0.146	1.33	1.18	0.15 (-0.074, 0.367)	0.273	1.41	1.26	0.15 (-0.055, 0.353)	0.162
Δ subcu ratio LM	1.40	1.63	-0.23 (-0.588, 0.123)	0.168	1.86	1.90	-0.04 (-0.614, 0.530)	0.735	1.36	1.40	-0.03 (-0.344, 0.278)	0.855
Δ subcu ratio hand	1.18	1.18	0.00 (-0.244, 0.237)	1.000	1.29	1.36	-0.07 (-0.244, 0.099)	0.685	1.26	1.14	0.12 (-0.120, 0.351)	0.208
Δ RC ratio UL	1.04	1.82	-0.78 (-2.002, 0.439)	0.146	1.18	1.52	-0.34 (-1.321, 0.644)	0.650	1.15	1.22	-0.07 (-0.627, 0.492)	0.295
Δ RC ratio ML	1.37	1.55	-0.18 (-1.609, 1.243)	0.244	1.70	0.91	0.79 (-0.151, 1.733)	0.027*	0.77	1.34	-0.57 (-1.321, 0.182)	0.194
Δ RC ratio LL	1.88	1.25	0.63 (-1.269, 2.524)	0.839	1.42	1.24	0.18 (-0.315, 0.683)	0.414	1.31	1.28	0.02 (-0.480, 0.525)	0.944
Δ RC ratio LM	1.04	1.02	0.02 (-0.652, 0.687)	0.839	1.28	1.25	0.03 (-0.951, 1.016)	0.839	0.81	0.89	-0.08 (-0.520, 0.369)	1.000
Δ RC ratio hand	0.81	2.65	-1.84 (-4.924, 1.247)	0.635	1.37	1.42	-0.05 (-0.650, 0.556)	0.839	1.68	1.06	0.62 (-0.699, 1.929)	0.268
Δ SWV ratio UL	1.40	1.04	0.36 (-0.120, 0.838)	0.216	0.96	0.92	0.03 (-0.194, 0.261)	0.442	0.99	1.24	-0.26 (-0.680, 0.169)	0.184
Δ SWV ratio UM	0.83	0.76	0.07 (-0.061, 0.193)	0.542	0.96	0.81	0.15 (-0.091, 0.388)	0.140	0.91	0.86	0.06 (-0.058, 0.171)	0.402
Δ SWV ratio LL	2.98	2.34	0.64 (0.079, 1.203)	0.033*	2.46	2.35	0.12 (-0.342, 0.575)	0.594	2.41	2.59	-0.18 (-1.023, 0.659)	0.889
Δ SWV ratio LM	2.43	2.07	0.36 (-0.290, 1.001)	0.191	1.94	2.19	-0.25 (-0.963, 0.454)	0.761	2.30	2.50	-0.20 (-0.895, 0.498)	0.780
Δ SWV ratio tx. site	2.25	1.71	0.54 (0.098, 0.980)	0.018*	1.73	1.38	0.35 (-0.265, 0.972)	0.149	1.90	1.62	0.28 (-0.245, 0.802)	0.244
ICG lymphography DB stage, median (IQR)												
ΔDB stage UL	3.64	3.21	0.43 (-0.063, 0.920)	0.089	3.33	3.33	0.00 (-0.209, 0.209)	1.000	3.50	3.29	0.21 (-0.301, 0.730)	0.586
Δ DB stage UM	3.57	3.36	0.21 (-0.032, 0.460)	0.149	3.33	3.20	0.13 (-0.153, 0.419)	1.000	3.64	3.43	0.21 (-0.249, 0.677)	1.000
Δ DB stage LL	3.29	3.71	-0.43 (-1.724, 0.867)	1.000	3.40	3.07	0.33 (-0.067, 0.734)	0.174	3.14	3.21	-0.07 (-0.345, 0.203)	0.773
Δ DB stage LM	3.43	3.07	0.36 (-0.009, 0.723)	0.089	3.07	3.20	0.00 (-0.419, 0.419)	1.000	3.00	2.93	0.07 (-0.284, 0.427)	1.000
Δ DB stage Hand	2.79	2.21	0.57 (-0.015, 1.158)	0.098	3.20	2.33	0.13 (-0.153, 0.419)	1.000	2.29	2.43	-0.14 (-0.353, 0.067)	0.346

Data are presented as median [interquartile range (IQR)] unless otherwise indicated. Within-group comparisons between pre- and post-intervention values were performed using paired t-tests or Wilcoxon signed-rank tests according to normality assessed via the Shapiro–Wilk test. Differences were calculated as pre-intervention values minus post-intervention values. The 95% confidence interval (CI) represents the estimated interval of the paired difference. The Pre and Post SWV values and corresponding difference signs were corrected in this revision to reflect the verified direction of change. Statistical significance was defined as $p<0.05$

MFBIA multifrequency bioelectrical impedance analysis, *Lo/Up* lower/upper extremity, *LIR* limb index ratio, *UL* upper lateral, *UM* upper medial, *LL* lower lateral, *LM* lower medial, *RC* resistance to compression, *SWV* shear wave velocity, *LLLT* low-level laser therapy, *ICG* indocyanine green, *DB* dermal back flow

promote lymphatic motoricity and lymph drainage to reduce fluid accumulation.

In contrast to these fluid-related outcomes, the change in SWV differed significantly among groups, with a greater reduction in Group A than in Group B, and significant within-group reductions in SWV were also observed at the lower-lateral and individually selected treatment sites in Group A. However, these findings were site-specific therefore, they should not be interpreted as evidence of overall superiority of 904 nm LLLT. Rather, they may indicate a local effect of 904 nm LLLT on tissue mechanical properties that is not directly attributable to measurable changes in tissue fluid or lymphatic drainage. Of particular interest, 904 nm, a wavelength that has been clinically used in BCRL protocols primarily intended to improve lymphatic drainage, demonstrated a more favorable SWV response than 650 nm when the irradiation target was shifted from proximal lymphatic structures to peripheral areas of chronic tissue change. This finding raises the possibility that the biological effects of 904 nm LLLT may not be restricted to lymphatic drainage and may also involve local tissue responses when the anatomical and pathological target of irradiation is modified. Nevertheless, because the significant between-group SWV difference was limited to a single measurement region and was demonstrated against Group B rather than consistently against sham, this finding should be considered exploratory and requires confirmation in larger studies.

LLLT has been reported in an in vivo studies to have an effect on macrophages and fibroblasts by decreasing pro-fibrotic transformation growth factor expression and type I collagen deposition by triggering metabolic energy [9]. Although, clinical trials targeting lymphedema have only explained this mechanism of LLLT on fibrotic tissue [25, 26], this study is the first trial to measure the variables related to the improvement of tissue softening in patient groups depending on different LLLT wavelength. In addition, as the optimal wavelength and energy dose that effectively penetrated the tissue in the lymphedema of the human body has not been proven, this result can be considered as a meaningful attempt to confirm the tissue softening effect of LLLT according to various wavelength and target areas in lymphedema group. Through the result of this study, is can be estimated that 904 nm LLLT may be more effective than 650 nm for fibrosis tissue breakdown and hardness reduction in the affected limb of lymphedema.

Recent systematic literature reviews on the application method of LLLT confirmed that most of them used wavelengths in the range of around 900 nm, and the group that received treatment showed a volume reduction effect. However, RCT reported in another study that applied a shorter wavelength also showed the effect of volume reduction. Otherwise, there still is lack of evidence confirming various outcomes in order to confirm the differences in effect according to different parameters of device and stratified treatment region as most of the study focused on circumference and pain improvement after single wavelength system of the device [27–29]. In particular, since TDC is proved to be able to evaluate the focal water component in lymphedema more sensitively than BIA [30]. In this study, although the overall TDC value improved in all groups, no specific group showed a significant difference after the treatment. Therefore, more clinical evidence is needed to derive the optimal protocols within a therapeutic range to prove the effectiveness of LLLT on lymphatic drainage and reducing tissue fluid accumulation.

In addition, extensive heterogeneity of protocols such as wavelength for tissue penetration, energy density, pulse duration, spot size of LLLT should be considered depending on the targeting tissue including skin fibrosis or lymph nodes, lymphatic vessels. As most of the clinical trials targeting the axillary lymph node, a previous study performed a single session of LLLT focusing improvement of skin fibrosis, showed only a temporary improvement in moisture content and tissue hardness [5]. Similarly, our study also focused on applying LLLT to the local area where fibrosis was clinically most severe, rather than the axillary LN, as the application site showing significant improvement of elasticity on the application site after 10 sessions of treatment. In the future, more clinical evidence is needed to derive appropriate parameters according to the severity of lymphedema and target tissue.

This study has several limitations. First, the modest sample size limited the statistical power to detect small between-group differences, and the site-specific SWV findings require confirmation in larger studies. Second, long-term follow-up was not performed; therefore, the durability of the observed changes in tissue mechanical properties remains uncertain. Third, most participants had relatively advanced lymphedema with extensive dermal backflow patterns, which may have limited the sensitivity of ICG lymphography for detecting subtle changes in lymphatic function. Finally, therapists could not be blinded to treatment allocation because device settings differed between groups, although participants and outcome assessors remained blinded. Future studies including larger populations, longer follow-up periods, and patients with earlier-stage lymphedema are warranted to confirm these findings.

Conclusion

In conclusion, neither 904 nm–650 nm LLLT produced significant additional changes in extracellular fluid-related measures or ICG lymphographic outcomes compared with

sham under the present treatment protocol. However, site-specific reductions in SWV were observed with 904 nm LLLT, including a greater reduction than with 650 nm at the upper-lateral region. These findings do not establish the overall superiority or antifibrotic efficacy of 904 nm LLLT but suggest that the response to LLLT in BCRL may depend on both wavelength and the anatomical and pathological tissue targeted. The use of an individualized, peripheral tissue-targeted protocol in the present study provides preliminary evidence supporting further investigation of LLLT not only as a lymphatic drainage-oriented therapy but also as a potential approach to chronic tissue changes in BCRL. Larger adequately powered studies with prespecified primary outcomes and longer-term follow-up are required to confirm these findings.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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